C9.2 Role of the master

The master must implement a *master exclusive monitor*, that is used to monitor the location used by an Exclusive sequence. This master exclusive monitor is used to determine if another master could have performed a store to the location during the Exclusive sequence, by monitoring the snoop transaction it receives.

When the master performs an Exclusive Load, the master exclusive monitor is set. The master exclusive monitor is reset when a snoop transaction is observed that indicates another master could perform a store to the location.

Note

In some implementations the cache line state is sufficient to provide the functionality of the master exclusive monitor. However, it is important that a line which is invalidated and made valid again by a mechanism such as prefetching, is not considered as having remained valid since the Exclusive Load.

C9.2.1 Exclusive Load

The master starts an Exclusive sequence with an Exclusive Load. The start of the Exclusive sequence must set the master exclusive monitor.

If the master does not hold a copy of the cache line then it must obtain a copy of the line using either a ReadClean or a ReadShared transaction.

An Exclusive Load transaction is a ReadClean or ReadShared transaction with the **ARLOCK** signal asserted. This indicates to the PoS exclusive monitor that the master is starting an Exclusive sequence.

It is recommended, but not required, that a master uses an Exclusive Load transaction, with **ARLOCK** asserted, if it is issuing a transaction at the start of Exclusive sequence. If a master does not use an Exclusive Load transaction, it is permitted to use a ReadClean or ReadShared transaction with **ARLOCK** de-asserted.

If the master holds a copy of the line in a Unique state, then issuing a transaction for the Exclusive Load is permitted but not recommended.

Note

This transaction is likely to cause an external memory access. It is also likely that informing the interconnect that an Exclusive sequence has started is unnecessary as there is no requirement to issue an Exclusive Store transaction to complete the sequence if the cache line remains in the Unique state.

If the master holds a copy of the line in a Shared state then issuing a transaction for the Exclusive Load is permitted, but not required.

Note

Issuing a transaction informs the interconnect that the master is performing an Exclusive sequence.

An Exclusive Load is expected to receive an EXOKAY response, which indicates that Exclusive accesses are supported at the address of the transaction. If Exclusive accesses are not supported then the transaction will receive an OKAY response.

C9.2.2 Exclusive Load to Exclusive Store

After the execution of an Exclusive Load a master will typically calculate a new value to store to the location before it attempts the Exclusive Store.

It is not required that a master always completes an Exclusive sequence. For example, the value obtained by the Exclusive Load can indicate that a semaphore is held by another master and therefore the value cannot be changed until the semaphore is released by the other master. Therefore, the Exclusive sequence can be restarted with no attempt to complete the current Exclusive sequence.

During the time between the Exclusive Load and the Exclusive Store the master exclusive monitor must monitor the location to determine whether another master might have performed a store to the location.

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C9.2.3 Exclusive Store

A master must not permit an Exclusive Store transaction to be in progress at the same time as any transaction that registers that it is performing an Exclusive sequence. The master must wait for any such transaction to complete before issuing an Exclusive Store transaction. The transactions that register a master is performing an Exclusive sequence are Exclusive Load transactions to any location, and Exclusive Store transactions to any location. These transactions are:

- ReadClean with **ARLOCK** asserted
- ReadShared with **ARLOCK** asserted
- CleanUnique with **ARLOCK** asserted.

When a master executes an Exclusive Store the following behavior is required:

- If the master exclusive monitor has been reset the Exclusive Store must fail and the master must not issue an Exclusive Store transaction. The master must restart the Exclusive sequence.

  Note

  Not issuing an Exclusive Store transaction when the master exclusive monitor has been reset prevents the issue of a transaction for an Exclusive Store that will eventually fail, and therefore avoids unnecessarily invalidating other copies of the line.

- If the line is held in a Unique state and the master exclusive monitor is set then the Exclusive Store has passed and the master can execute the Exclusive Store without issuing a transaction.

- If the line is held in a Shared state and the master exclusive monitor is set then the master must issue a transaction to perform the Exclusive Store. This check of the master exclusive monitor must only occur after any other transactions that register a master is performing an Exclusive sequence have completed. A CleanUnique transaction with **ARLOCK** asserted must be used. The master exclusive monitor must continue to operate during this transaction. The transaction will respond with an OKAY or EXOKAY response.

If the transaction receives an EXOKAY response then this indicates that the transaction has passed and been propagated to invalidate all other copies of the line. After an Exclusive transaction completes with an EXOKAY response the master must again check the master exclusive monitor:

- If the master exclusive monitor has been reset, it indicates that a non-Exclusive Store has occurred to the cache between the point that the Exclusive Store transaction was issued and the point that it completed. The Exclusive Store must fail and the Exclusive sequence restarted.

- If the master exclusive monitor is set then the Exclusive Store has passed and the store is performed.

If the transaction receives an OKAY response then this indicates another master has been permitted to progress a transaction associated with an Exclusive Store. The transaction associated with the Exclusive Store from this master has failed and has not propagated to other masters in the system. When an Exclusive transaction completes with an OKAY response the following options exist:

- the master can fail the Exclusive Store and restart the Exclusive sequence without checking the state of the line when the access completed

- the master can check the master exclusive monitor:
  - if the master exclusive monitor has been reset then the master must fail the Exclusive Store and restart the Exclusive sequence
  - if the master exclusive monitor is set then the master can repeat the Exclusive Store transaction.

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